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Determine the 8-bit two's complement of the following numbers: -1_{10}, -17_{10}, -128_{10}.

Convert the following numbers to decimal:

$$01111111_{2C}, \qquad 11001110_{2C}, \qquad 10001000_{2C}.$$

3) Determine the two's complement representations of the numbers $-63A_{16}$ and $-8AC_{16}$.

4) Convert the following numbers to ASCII codes:

$$\text{a) } 1_{10}, \qquad \text{b) } 107_{10}, \qquad \text{c) } 1000010_2.$$

5) Use ASCII code to translate the following expressions:

$$X = \cos(.7)$$
$$Y = 256 * X$$
$$\text{PRINT "X=", X; "Y=", Y}$$

6) a) Represent the following numbers in single-precision IEEE-754 standard: 2.75_{10}, $-417\ 680_{10}$.

b) Find the decimal number corresponding to each of the following single-precision IEEE-754 representations:

$$0\ 10001010\ 01110111000110000000000_{IEE754}$$
$$00000000\ 00000000000000000000001_{IEEE754}$$

EXERCISE 1.3.– Gray code/binary number conversion

1) Convert the following binary numbers to Gray code:

$$\text{a) } 11011_2 \qquad \text{b) } 101101_2 \qquad \text{c) } 11000110_2$$

2) Convert each of the following Gray code to binary number:

$$\text{a) } 1010_{\text{Gray}} \qquad \text{b) } 00010_{\text{Gray}} \qquad \text{c) } 11000010001_{\text{Gray}}$$

EXERCISE 1.4.– Correction code

1) Data are to be transmitted, coded by the CRC method, whose polynomial generator is:

$$G(x) = x^3 + x + 1$$

Determine the bits of the message to be transmitted if the initial message is 101101.

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What can we say of the transmission if the received message is 100101 011?

2) Using the CRC coding method, represent the message to be transmitted in the case where the initial information is 1011001 and the polynomial generator is of the form:

$$G(x) = x^4 + x + 1$$

1.15. Solutions

SOLUTION 1.1.– Conversions

1) Conversions to binary representation

a) $37_{10} = 100101_2$

b) $15_{10} = 1111_2$

c) $187_{10} = 10111011_2$

d) $2\,014_{10} = 11111011110_2$

e) $2\,016_{10} = 2^{11} - 2^5 = 11111100000_2$

f) $2.75_{10} = 10.11_2$

g) $25.25_{10} = 11001.01$

h) $243.3125_{10} = 11110011.0101_2$

i) $0.0625_{10} = 0.0001_2$

j) $62_8 = 110010_2$

k) $277_8 = 10111111_2$

l) $12.6_8 = 1010.11_2$

m) $476.35_8 = 100111110.011101_2$

n) $92_{16} = 10010010_2$

o) $37FD_{16} = 11011111111101_2$

p) $7FF_{16} = 11111111111_2$

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q) $1A6_{16} = 110100110_2$

r) $2C0_{16} = 1111000000_2$

s) $1F.C_{16} = 11111.11_2$

t) $9.F_{16} = 1001.1111_2$

u) $A7, EC_{16} = 10100111.111011_2$

2) Conversion to decimal representation

a) $10110_2 = 22_{10}$

b) $10001_2 = 17_{10}$

c) $10001101_2 = 141_{10}$

d) $100100001001_2 = 2313_{10}$

e) $1111010111_2 = 983_{10}$

f) $1011.101_2 = 11.625_{10}$

g) $10011011001.10110_2 = 1241.6875_{10}$

h) $30_8 = 36_{10}$

i) $115_8 = 77_{10}$

j) $55.4_8 = 45.5_{10}$

k) $270.54_8 = 184.6875_{10}$

l) $356_{16} = 854_{10}$

m) $2AF_{16} = 687_{10}$

n) $2C1_{16} = 705_{10}$

o) $10FF_{16} = 4351_{10}$

p) $1FCFA_{16} = 130298_{10}$

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q) $DADA.C_{16} = 56026.75_{10}$

r) $F.4_{16} = 15.25_{10}$

s) $EBA.C_{16} = 3770.75_{10}$

3) Conversion to hexadecimal representation

a) $320_{10} = 140_{16}$

b) $6\ 861_{10} = 1ACD_{16}$

c) $65\ 535_{10} = 16^4 - 1 = FFFF_{16}$

d) $100_8 = 40_{16}$

e) $62.4_8 = 32.8_{16}$

f) $500.25_8 = 140.54_{16}$

g) $10001101_2 = 8D_{16}$

h) $1001000110100011110_2 = 48D1E_{16}$

i) $10000.1_2 = 10.8_{16}$

j) $1000000.0000111_2 = 40.0E_{16}$

k) $1000111001.01_2 = 239.4_{16}$

4) BCD – Decimal conversion

a) $0001\ 1000\ 0100_{BCD} = 184_{10}$

b) $0100\ 1001\ 0010_{BCD} = 492_{10}$

c) $1001\ 0111\ 0101\ 0010_{BCD} = 9\ 752_{10}$

d) $0111\ 0111\ 0111\ 0101_{BCD} = 7\ 775_{10}$

5) How many bits are required for the binary representations of the decimal numbers from 0 to 511?

With k bits, only the decimal numbers from 0 to $2^k - 1$ can be represented. Thus:

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$$2^k - 1 = 511 \quad \text{and} \quad k = \log(512)/\log(2) = 9$$

6) What is the largest number that can be represented in 16 bit binary numeration?

The largest number that can be represented in 16 bits binary numeration system is $2^{16} - 1 = 65.535$.

7) a) Determine the binary representation of the decimal number 10.05 with an absolute error equal to 0.005.

To ensure that the difference between the represented value and the decimal number 10.05 remains less than or equal to 0.005, the required number of bits, *n*, must be determined based on the following relationship:

$$2^{-n} \leq 0.005$$

that is:

$$n \geq \log(1/0.005)/\log(2) = 7.64 \simeq 8$$

and finally we have:

$$10.05_{10} = 1010.00001100_2$$

b) Represent the decimal number 0.452 in binary with a relative error of 0.1%.

The desired value of the absolute error is $0.001 \times 0.452 = 0.000452$.

The binary representation must possess a number of bits, *n*, that satisfies the following equation:

$$2^{-n} < 0.000452$$

that is:

$$n > \log(1/0.000452)/\log(2) = 11.11 \simeq 12$$

Thus:

$$0.452_2 = 0.0111001110110110_2$$

SOLUTION 1.2.– Representation of numbers and data

1) Let X be an unsigned n-bit integer. Verify that $2^n - X$ represents the 2C of $-X$.

Let $X = X_{n-1}X_{n-2} \cdots X_1X_0$ be the n-bit binary representation of a positive number. We can write:

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$$2^n - X = (2^n - 1) - X + 1$$

where:

$$2^n - 1 = \underbrace{111\cdots11}_{n \text{ bits}}{}_2$$

The following subtraction operation:

$$\underbrace{111\cdots11}_{n \text{ bits}}{}_2 - X_{n-1}X_{n-2}\cdots X_1X_0$$

is equal to the inversion of the logical level of the bits of X. Thus:

$$2^n - X = (2^n - 1) - X + 1 = \overline{X} + 1$$

is the 2C representation of $-X$.

2) Determine the 8-bit 2C of the following numbers: -1_{10}, -17_{10}, -128_{10}.

For each of the numbers, we can obtain:

$$\begin{aligned} 1_{10} &= 00000001_2 \\ &\quad 11111110_{1C} + 1 = 11111111_{2C} = -1_{10} \end{aligned}$$

$$\begin{aligned} 17_{10} &= 00010001_2 \\ &\quad 11101110_{1C} + 1 = 11101111_{2C} = -17_{10} \\ 128_{10} &= 10000000_2 \\ &\quad 01111111_{1C} + 1 = 10000000_{2C} = -128_{10} \end{aligned}$$

3) Convert the following numbers to decimal representation: 01111111_{2C}, 11001110_{2C} and 10001000_{2C}.

The conversions are carried out as follows:

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$$01111111_{2C} = 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 127_{10}$$
$$11001110_{2C} = -2^7 + 2^6 + 2^3 + 2^2 + 2^1 = -50_{10}$$
$$10001000_{2C} = -2^7 + 2^3 = -120_{10}$$

4) Determine the 2C of the numbers $-63A_{16}$ and $-8AC_{16}$.

We can proceed as follows:

$$\begin{array}{rl} 63A_{16} = & 011000111010_2 \\ & 100111000101_{1C} \\ + & \qquad\qquad\qquad 1 \\ \hline = & 100111000110_{2C} = -63A_{16} = 9C6_{16} \end{array}$$

$$\begin{array}{rl} 8AC_{16} = & 0000100010101100_2 \\ & 1111011101010011_{1C} \\ + & \qquad\qquad\qquad\qquad 1 \\ \hline = & 1111011101010100_{2C} = -8AC_{16} = F754_{16} \end{array}$$

5) Convert the following numbers to ASCII code:

a) 1_{10} b) 107_{10} c) 1000010_2

The corresponding ASCII codes are as follows:

a) 1_{10} : SOH (start of heading) b) 107_{10} : k
c) $1000010_2 = 66_{10}$: B

6) Use ASCII code to translate the following expressions:

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$$X = \cos(.7)$$
$$Y = 256 * X$$
PRINT "X=", X; "Y=", Y

	ASCII code	
Symbol	Dec.	Hex.
X	88	58
=	61	3D
c	99	63
o	111	6F
s	115	73
(	40	28
.	46	2E
7	55	37
)	41	29
Y	89	59
=	61	3D
2	50	32
5	53	35
6	54	36
*	42	2A
X	88	58

	ASCII code	
Symbol	Dec.	Hex.
P	80	50
R	82	52
I	73	49
N	78	4E
T	84	54
Espace	32	20
"	34	22
X	88	58
=	61	3D
"	34	22
,	44	2C
X	88	58
"	34	22
Y	89	59
=	61	3D
"	34	22
,	44	2C
Y	89	59

7) a) Represent the following numbers in single-precision IEEE-754 standard: 2.75_{10} and $-417\,680_{10}$.

The binary conversion of 2.75_{10} yields:

$$2.75_{10} = 10.11_2$$

The normalized form is written as:

$$10.11_2 = 1.011_2 \times 2^1$$

We thus have:

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– the mantissa $M = 1.f$ (f represents the 23 bit fractional part):

$$f = 01100000000000000000000$$

– the exponent of 8 bits:

$$E_b = E + b = 1_{10} + 127_{10} = 128_{10} = 10000000_2$$

– the sign bit:

$S = 0$ (in the case of a positive number)

From which:

$$2.75_{10} = 0\ 10000000\ 01100000000000000000000$$

For the decimal number −417 680, we obtain a binary representation of the following form:

$$417\ 680_{10} = 1100101111110010000_2$$

The corresponding normalized form can be expressed as:

$$1100101111110010000_2 = 1.100101111110010000_2 \times 2^{18}$$

We thus have:

– the mantissa $M = 1.f$ (f representing the 23 bit fractional part):

$$f = 10010111111001000000000$$

– the exponent of 8 bits:

$$E_b = E + b = 18_{10} + 127_{10} = 145_{10} = 10010001_2$$

– the sign bit:

$S = 1$ (in the case of a negative number)

And finally:

$$-417\ 680_{10} = 1\ 10010001\ 10010111111001000000000_{IEEE754}$$

b) Find the decimal number corresponding to each of the following single-precision IEEE-754 representations:

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$$0\ 10001010\ 01110111000110000000000_{IEE754}$$

$$1\ 00000000\ 00000000000000000000001_{IEEE754}$$

For N_{10} = 0 10001010 01110111000110000000000$_{IEE754}$, we have:

– the sign bit:

S = 0 (in the case of a positive number)

– the exponent of 8 bits:

$$E_b = 10001010_2 = 138_{10} \quad \text{(normalized number)}$$

and:

$$E = E_b - b = 138 - 127 = 11_{10}$$

– the mantissa M of 23 bits:

$$\begin{aligned} M &= 1.f \\ &= 1.01110111000110000000000_2 \\ &= 2^0 + 2^{-2} + 2^{-3} + 2^{-4} + 2^{-6} + 2^{-7} + 2^{-8} + 2^{-12} + 2^{-13} \\ &= 1.4652099609375_{10} \end{aligned}$$

Hence:

$$N_{10} = (-1)^S M \times 2^E = 1.4652099609375 \times 2^{11} = 3000.75_{10}$$

In the case of N_{10} = 1 00000000 00000000000000000000001$_{IEEE754}$, we obtain:

the sign bit:

$S = 1$ (for a negative number)

the exponent of 8 bits:

$$E_b = 00000000_2 = 0_{10} \quad \text{(denormalized number)}$$

and:

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$$E = E_b - b + 1 = 0 - 127 + 1 = -126_{10}$$

the mantissa M of 23 bits:

$$\begin{aligned} M &= 0.f \\ &= 0.00000000000000000001 \\ &= 2^{-23} \end{aligned}$$

and finally:

$$N_{10} = (-1)^S M \times 2^E = -2^{-23} \times 2^{-126} = -2^{-149} = -1.4 \times 10^{-45}$$

SOLUTION 1.3.– Gray code/binary number conversion

1) Binary–Gray code conversion:

$$\text{a) } 11011_2 = 10110_{\text{Gray}} \qquad \text{b) } 101101_2 = 111011_{\text{Gray}}$$

$$\text{c) } 11000110_2 = 10100101_{\text{Gray}}$$

2) Gray code–binary number conversion:

$$\text{a) } 1010_{\text{Gray}} = 1100_2 \qquad \text{b) } 00010_{\text{Gray}} = 00011_2$$

$$\text{c) } 11000010001_{\text{Gray}} = 10000011110_2$$

SOLUTION 1.4.– Correction code

1) The principle of the CRC method consists of processing words and codes as binary polynomials.

– *Coding*

In this case, the polynomial generator is of the degree 3 (r = 3) and is written as:

$$G(x) = x^3 + x + 1$$

The correspondence between the initial message and a polynomial form is established as follows:

$$101101 \quad \leftrightarrow \quad I(x) = 1 \cdot x^5 + 0 \cdot x^4 + 1 \cdot x^3 + 1 \cdot x^2 + 0 \cdot x^1 + 1$$

The polynomial associated with the initial message is reduced to:

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$$I(x) = x^5 + x^3 + x^2 + 1$$

We have:

$$I(x)x^3 = (x^5 + x^3 + x^2 + 1)x^3 = x^8 + x^6 + x^5 + x^3$$

The division of the polynomial $I(x)x^3$ by $G(x)$ is performed as follows:

$$\begin{array}{l|ll}
x^8 + x^6 + x^5 + x^3 & x^3 + x + 1 & \\
\cline{2-2}
\underline{x^8 + x^6 + x^5} & x^5 + 1 & \leftarrow \text{Quotient} \\
\qquad\qquad x^3 & & \\
\qquad\qquad \underline{x^3 + x + 1} & & \\
\qquad\qquad\qquad x + 1 & \leftarrow \text{Remainder} &
\end{array}$$

The message to be transmitted, $T(x)$, is obtained by concatenating the bits of the initial message, $I(x)$, and the r bits of the division remainder. That is:

$$101101\ 011$$

It must be noted that the division is carried out using the addition modulo 2 for the coefficients of the polynomials (0 + 0 = 0, 1 + 0 = 1, 1 + 1 = 0).

– *Verification*

For an error-free transmission, the remainder of the division of the received message, $T(x)$, by the polynomial generator, $G(x)$, must be equal to 0.

The polynomial associated with the received message is of the form:

$$101101\ 011 \quad \leftrightarrow \quad x^8 + x^6 + x^5 + x^3 + x + 1$$

The division is executed as follows:

$$\begin{array}{l|ll}
x^8 + x^6 + x^5 + x^3 + x + 1 & x^3 + x + 1 & \\
\cline{2-2}
\underline{x^8 + x^6 + x^5} & x^5 + 1 & \leftarrow \text{Quotient} \\
\qquad\qquad x^3 + x + 1 & & \\
\qquad\qquad \underline{x^3 + x + 1} & & \\
\qquad\qquad\qquad 0 & \leftarrow \text{Remainder} &
\end{array}$$

The remainder being 0, this is an error-free transmission.

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Considering the other message that is received, we can obtain the following polynomial:

$$100101\ 011 \quad \leftrightarrow \quad x^8 + x^5 + x^3 + x + 1$$

The division is carried out as follows:

$$\begin{array}{l|l}
x^8 + \quad x^5 + \quad x^3 + \quad x + 1 & x^3 + x + 1 \\
\cline{2-2}
x^8 + x^6 + x^5 & x^5 + x^3 + x \leftarrow \text{Quotient} \\
\hline
\quad x^6 + \quad x^3 + \quad x + 1 & \\
\quad x^6 + \quad x^4 + x^3 & \\
\hline
\quad x^4 + \quad x + 1 & \\
\quad x^4 + \quad x^2 + x & \\
\hline
\quad x^2 + \quad 1 & \leftarrow \text{Remainder}
\end{array}$$

The remainder of the division is not equal to 0, which is the feature of a received message with errors.

2) Coding of the message to be transmitted when the polynomial generator is $G(x) = x^4 + x + 1$ and the initial message is 1011001.

The initial message can be associated with the polynomial, $I(x)$, given by:

$$I(x) = x^6 + x^4 + x^3 + 1$$

The multiplication of $I(x)$ by x^4 results in:

$$I(x)x^4 = (x^6 + x^4 + x^3 + 1)x^4 = x^{10} + x^8 + x^7 + x^4$$

The division is performed as follows:

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$$
\begin{array}{l|l}
x^{10} + x^8 + x^7 + \qquad\qquad x^4 & x^4 + x + 1 \\
\cline{2-2}
x^{10} + \qquad x^7 + x^6 & x^6 + x^4 + x^2 + x \leftarrow \text{Quotient} \\
\cline{1-1}
\qquad x^8 + \qquad x^6 + \qquad x^4 & \\
\qquad x^8 + \qquad\qquad x^5 + x^4 & \\
\cline{1-1}
\qquad\qquad x^6 + x^5 & \\
\qquad\qquad x^6 + \qquad\qquad x^3 + x^2 & \\
\cline{1-1}
\qquad\qquad\qquad x^5 + \quad x^3 + x^2 & \\
\qquad\qquad\qquad x^5 + \qquad\qquad x^2 + x & \\
\cline{1-1}
\qquad\qquad\qquad\qquad\qquad x^3 + \qquad x & \leftarrow \text{Remainder}
\end{array}
$$

Concatenating the bits of the initial message, $I(x)$, and the four bits of the remainder of the division, the message to be transmitted, $T(x)$, takes the following form:

$$1011001\ 1010$$

[1] IEEE: Institute of Electrical and Electronics Engineers.

[2] CRC: cyclic redundancy check.

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